Taguchi Methods

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Introduction

Taguchi methods (1950s-60s) apply orthogonal-array experiments to optimise product and process design for minimum variability. Central ideas: loss functions defining quality as deviation from target, signal-to-noise (SNR) ratios summarising mean and variance jointly, and compact orthogonal arrays for efficient screening.

Prerequisites

Fractional factorial designs; signal-to-noise.

Theory

Quality loss $L(y) = k(y - \tau)^2$ for target τ – quality decreases quadratically with deviation.

SNR: - Smaller-is-better: $\text{SNR} = -10\log_{10}(\sum y_i^2/n)$. - Larger-is-better: $\text{SNR} = -10\log_{10}(\sum 1/y_i^2/n)$. - Nominal-is-best: $\text{SNR} = 10\log_{10}(\bar{y}^2/s^2)$.

Taguchi arrays (L4, L8, L9, L16) are Plackett-Burman-style orthogonal arrays with Japanese quality-engineering conventions.

Assumptions

Main effects dominate (sparsity); variance reduction via control factor choice (robust design).

R Implementation

```r
library(qualityTools)

set.seed(2026)
# L8 array (2-level, 7 factors)
ta <- taguchiDesign("L8_2^7")
summary(ta)

# Simulate responses
ta$y <- rnorm(8, mean = 10 + 2 * as.numeric(ta$A) +
                         1.5 * as.numeric(ta$B), sd = 0.5)
effectPlot(ta, response = "y",
           main = "Taguchi main-effects plot (L8)")
```

Output & Results

L8 orthogonal array and main-effects plot showing the average response at each factor's two levels.

Interpretation

"The L8 design screened 7 factors in 8 runs; the main-effects plot identified factors A and B as dominant, with the optimal settings maximising (or minimising) the target response."

Practical Tips

- Taguchi arrays are essentially Plackett-Burman designs; modern DoE uses statistical fractional factorials with aliasing awareness.
- SNR metrics conflate mean and variance; modern practice separates them via dual-response surface methods.
- Taguchi's three-stage design (system, parameter, tolerance) is a useful framework; the specific SNR tools are debated.
- Orthogonal arrays are efficient for screening; interactions must be explicitly assigned to specific columns.
- For manufacturing and quality engineering, Taguchi remains a widely-used vocabulary; modern methods supersede the specifics.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Observational designs and STROBE
- Randomised controlled trials
- Adaptive, non-inferiority, equivalence trials
- Bench and translational design
- MCAR, MAR, MNAR
- Multiple imputation with mice
- Linear mixed models with lme4
- GLMMs and GEE
- DAGs with dagitty and ggdag
- Propensity scores and IPTW
- G-methods, IV, DiD, RDD; heterogeneity of treatment effect
- SIR and SEIR models with deSolve
- Pre-registration and statistical analysis plans